Assignment_5

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Question 1

Apply hierarchical clustering using Euclidean distance and compare single, complete, average, and Ward's linkage. Choose the best method.

```r
library(cluster)

# Read data
cereals <- read.csv("Cereals.csv")

# Data preprocessing: remove missing values, keep numeric variables, standardize them
cereals_clean <- na.omit(cereals)
cereals_num <- cereals_clean[, sapply(cereals_clean, is.numeric)]
cereals_norm <- scale(cereals_num)

# Compare linkage methods
agnes_single   <- agnes(cereals_norm, method = "single")
agnes_complete <- agnes(cereals_norm, method = "complete")
agnes_average  <- agnes(cereals_norm, method = "average")
agnes_ward     <- agnes(cereals_norm, method = "ward")

# Show agglomerative coefficients
agnes_single$ac
```

```
## [1] 0.6067859
```

```r
agnes_complete$ac
```

```
## [1] 0.8353712
```

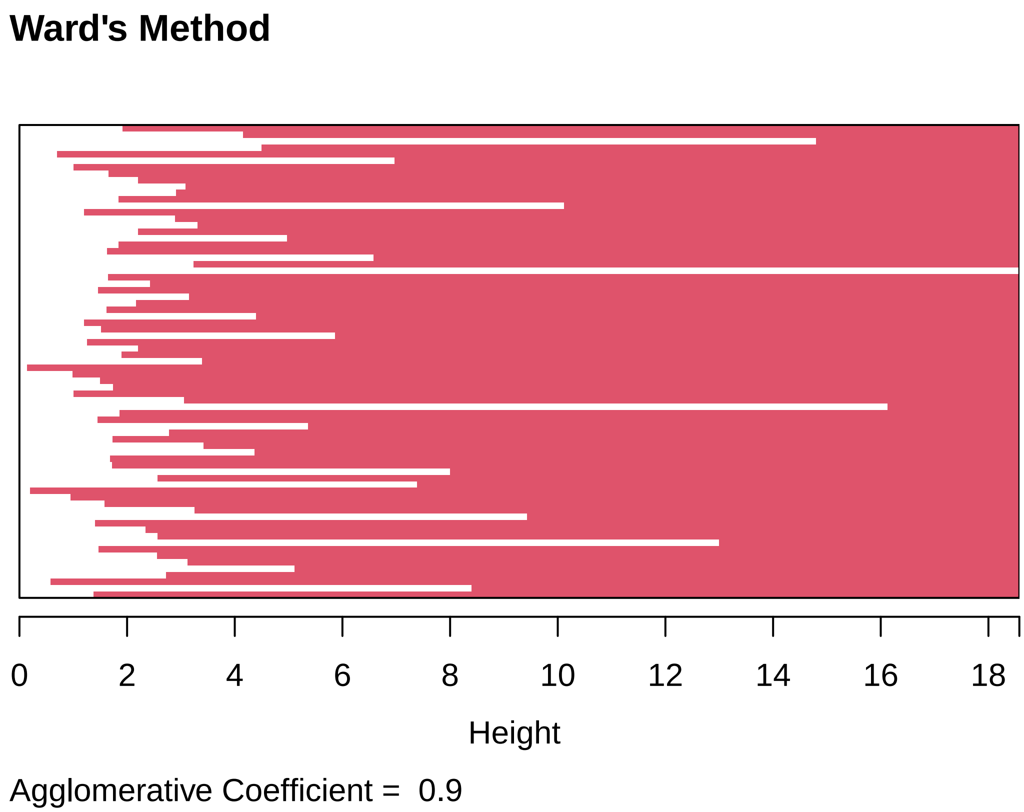
```r
agnes_average$ac
```

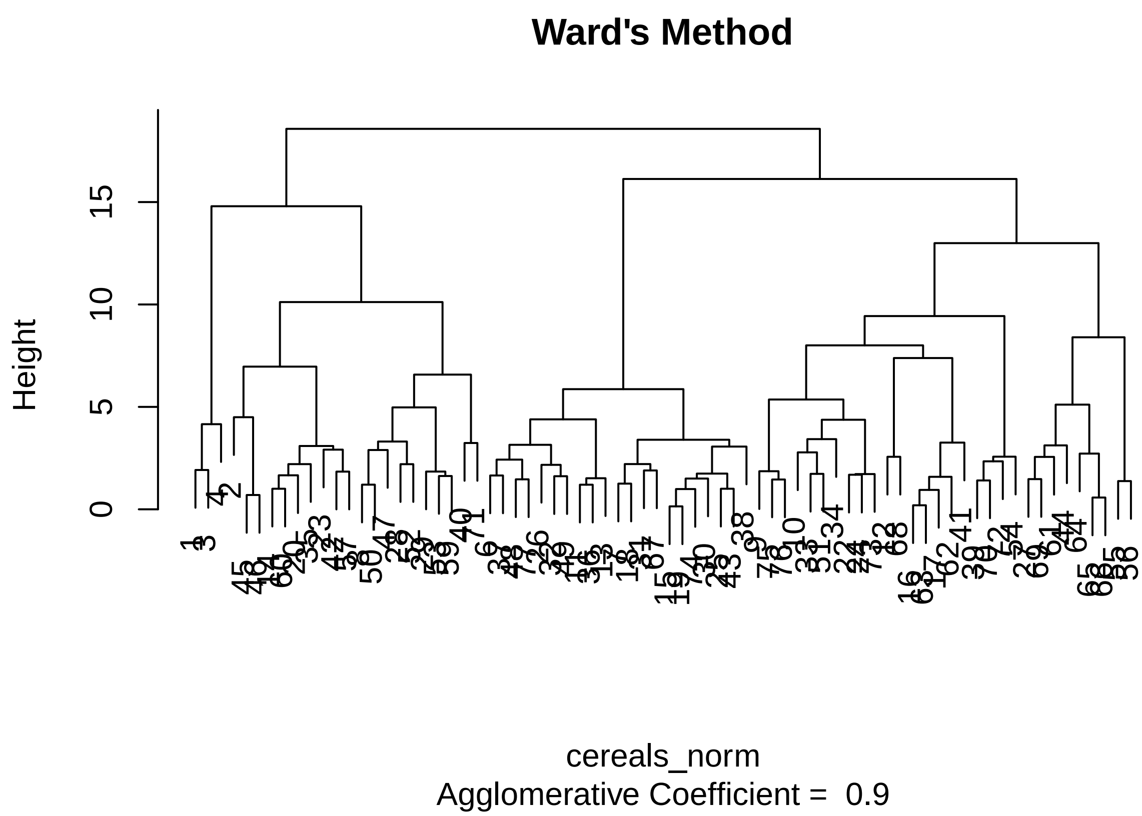
```
## [1] 0.7766075
```

```r
agnes_ward$ac
```

```
## [1] 0.9046042
```

```r
# Plot Ward's method
plot(agnes_ward, main = "Ward’s Method")
```





I compared four linkage methods using the `agnes()` function. The agglomerative coefficient was highest for **Ward's method**, which means it created the cleanest and most well-defined clusters. Ward's method tries to keep the clusters tight, and based on both the numbers and the dendrogram, it clearly performed the best.

Question 2

How many clusters would you choose?

Looking at the dendrogram for Ward's method, the biggest jump in height happens right before the data split into **four main branches**, so I chose **4 clusters**.

Question 3

Comment on the structure of the clusters and their stability.

```r
# Partition the dataset into A and B
set.seed(123)
index <- sample(1:nrow(cereals_norm), nrow(cereals_norm) / 2)
```

```r
A <- cereals_norm[index, ]
B <- cereals_norm[-index, ]

# Cluster A
agnes_A <- agnes(A, method = "ward")
clusters_A <- cutree(agnes_A, k = 4)
table(clusters_A)
```

```
## clusters_A
##  1  2  3  4
## 12 11 10  4
```

```r
# Compute centroids for A
centroids_A <- aggregate(A, by = list(cluster = clusters_A), mean)
centroids_mat <- as.matrix(centroids_A[, -1])

# Assign records in B to nearest centroid
dist_B_to_centroids <- as.matrix(dist(rbind(centroids_mat, B)))[-(1:4), 1:4]
assigned_B <- apply(dist_B_to_centroids, 1, which.min)
table(assigned_B)
```

```
## assigned_B
##  1  2  3  4
## 13  6 11  7
```

```r
# Full-data clusters for comparison
full_clusters <- cutree(agnes_ward, k = 4)
full_clusters_B <- full_clusters[-index]

table(assigned_B, full_clusters_B)
```

```
##           full_clusters_B
## assigned_B  1  2  3  4
##          1  3  1  0  9
##          2  0  6  0  0
##          3  0  1 10  0
##          4  0  0  0  7
```

The clusters in Partition A were fairly balanced, and when I checked stability by assigning Partition B cereals to the nearest centroid, most cereals matched their full-data cluster. Clusters 2, 3, and 4 were very stable. Cluster 1 had a little overlap with Cluster 4, which suggests those cereals sit close together. Overall, the clustering looked stable.

Question 4

Identify the "healthy cereal" cluster.

Cluster **1** came out as the "healthy" group. These cereals had lower sugars and calories, and higher protein and fiber, which makes them good options for elementary school cafeterias.

Question 5

Should the data be normalized?

Yes. The variables are all on different scales, and without normalization, the variables with large ranges (like sodium) would dominate the clustering. Normalizing ensures all the nutritional variables contribute fairly.